

EEE/ECE-103IL Lab Report 3 (Part-A)

[Conditional Statement]

1. Write a C program that takes an input voltage level (in volts) from user and classifies it based on the following conditions and prints the corresponding message:
 - a. 0V to 5V: “Logic Level Voltage”
 - b. More than 5V and up to 230V: “AC Mains Voltage”
 - c. More than 230V: “High Voltage – Dangerous!”

2. In a three-phase power system, the voltages of three phases (V_a , V_b , V_c) are given by the user. Write a program that checks the following conditions:
 - a. If all three phases are within the range of 220V to 240V, print “System is balanced.”
 - b. If any one phase is below 220V, print “Phase undervoltage detected.”
 - c. If any one phase is above 240V, print “Phase overvoltage detected.”
 - d. If one of the three phases is missing (i.e., voltage = 0V), print “Phase failure detected.”

3. A motor is protected by a simple logic-based system. The user will input the motor temperature (in °C) and supply voltage (in volts), write a program to:
 - a. If the temperature is below 70°C and the voltage is between 210V-230V, print “Motor running normally.”
 - b. If the temperature is above 70°C but below 90°C, print “Warning: Motor overheating!”
 - c. If the temperature exceeds 90°C, print “Motor shutdown – Overheated!”
 - d. If the voltage is below 210V or above 250V, print “Voltage abnormal – Motor shutdown!”

4. Write a program that takes the input voltage (in volts) and load resistance R (in Ω) and prints the appropriate wire gauge based on current rating: [Hint: Determine current first]
 - a. If current $\leq 5A$: “Use 1mm² wire.”
 - b. If $5A < \text{current} \leq 15A$: “Use 2.5mm² wire.”
 - c. If $15A < \text{current} \leq 30A$: “Use 4mm² wire.”
 - d. If current $> 30A$: “Use 6mm² or higher gauge wire.”

EEE-103IL: Computer Programming Laboratory

Lab 3(B): Loop Related Problems

Q1.

Write a C program to count and display all the numbers between **a** and **b** (where $a < b$) that are divisible by both 3 and 5. Both **a** and **b** should be taken as user inputs.

Sample Output:

```
Input lower limit (a): 10
```

```
Input upper limit (b): 50
```

```
Numbers divisible by both 3 and 5 are: 15 30 45
```

```
Total count = 3
```

Q2.

Write a C program to display the value of n^n for all numbers from 1 up to **n**.

Sample Output:

```
Input number of terms: 5
```

```
1^1 = 1
```

```
2^2 = 4
```

```
3^3 = 9
```

```
4^4 = 16
```

```
5^5 = 25
```

Q3.

Write a C program to print an $n \times n$ square pattern using the * symbol. The value of **n** will be taken as user input.

Sample Output:

```
Input number of rows (n): 4
```

```
* * * *
```

```
* * * *
```

```
* * * *
```

```
* * * *
```

Q4.

Write a C program to print a right-aligned right-angled triangle using sequentially increasing numbers in each row. The value of **n** (number of rows) will be taken as user

input.

Sample Output:

Input number of rows (n): 5

1

12

123

1234

12345